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# Value in Vogue ( *VinV* ) – Applied Value Behavior Under Macro Regimes

*Downstream Proof of Concept  
(Non-Forecasting, Descriptive)*



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## Purpose & Boundary

**VinV** is a **downstream applied research layer, explicitly separated from FT-GMI & free of forecasting or allocation claims.**

02

## Existing Data Artifacts

Summarizes the **canonical equity-macro panel, returns integration, & evaluation machinery** already built.

03

## Regime Conditioning

All analysis is **segmented by macro regimes** rather than pooled across **full samples.**

04

## Behavior Under Stress

**Descriptively compares VinV with related benchmarks across drawdown & recovery windows.**

05

## Observed Structure

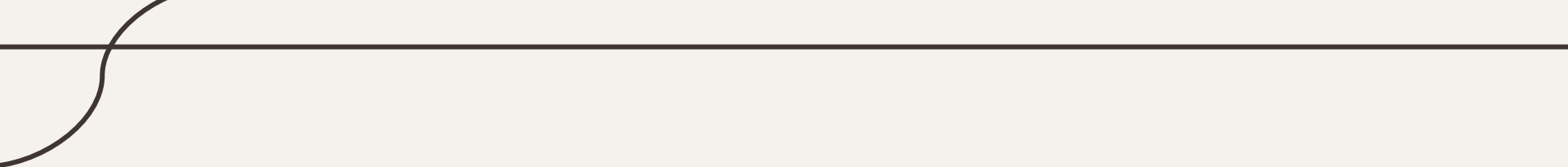
Highlights **dispersion, volatility, persistence, & recovery slope** as **behavioral properties, not signals.**

06

## Completion Path

States what **VinV is not yet & how it completes** after **FT-GMI is finalized without contaminating diagnostics.**

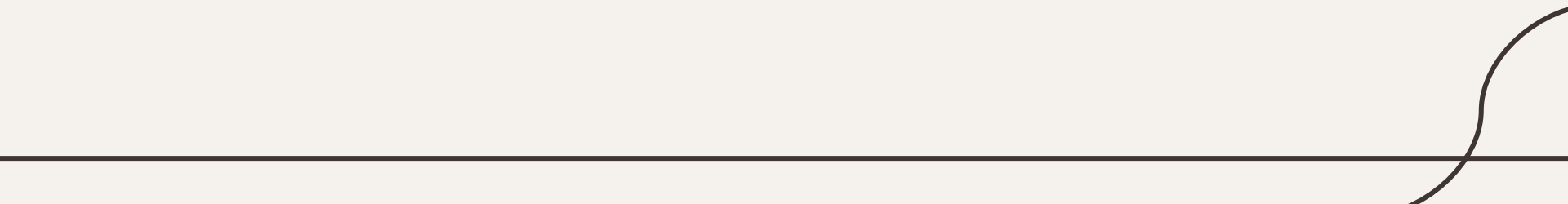
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# 01

# Purpose & Boundary

*Defines VinV's scope, intent, & explicit non-claims to enforce clear analytical boundaries from the outset.*



# System Context & Governance Boundary –

*Layered macro-financial research architecture separating data, diagnostics, interpretation, & application.*



## *the Spine* (Foundational Layer)

- **Canonical macro-financial data fusion & normalization,**
- **G20-scope macro-financial inputs** across rates, FX, inflation, credit, commodities, equities, & real economy metrics,
- **Deterministic feature contracts with locked semantics,**
- **Canonical raw data stored** in governed **cloud object storage (Cloudflare R2);** **read/write access restricted to the Spine layer only,** &
- Infrastructure for **reproducibility, auditability, & scale.**



## Downstream Diagnostic & Application Layers

- **FT-GMI** — regime-aware macro-financial diagnostics (*read-only*),
- **VinV** — U.S. equity behavior conditioned on macro regimes, &
- **the OracleChambers** — interpretive narrative layer (*non-authoritative*).

**AI/ML tooling** constrained to **monitoring, explanation, & validation support** — operating **strictly downstream** ("*living off the exhaust*") in **the OracleChambers**.

# FT-GMI Introduction –

*Regime-aware macro-financial diagnostics layer*

## What is FT-GMI?

- It measures **structural macro-financial stress within the G20**, not outcomes,
- Its **outputs define regimes & stress windows** used by VinV,
- FT-GMI is **read-only & never altered by downstream analysis**,
- **FT-GMI consumes canonical macro-financial data produced by the Spine (fusion & normalization layer)**, &
- **the Spine** is the **canonical fusion, normalization, & governance layer** supplying all macro-financial inputs to FT-GMI.



## Relationship FT-GMI & VinV

- **VinV** is a **U.S.-focused downstream layer** within the **FT-GMI layer**,
- **VinV conditions equity behavior** on the **macro context** provided by **FT-GMI**, &
- The dependency is strictly one-way ( $FT-GMI \rightarrow VinV$ ).

# VinV: Why This Exists

## Purpose Statement

VinV is a **downstream applied research layer** designed to *describe how value-oriented equity constructions behave under different macro regimes.*

## What VinV Does

- Examines **historical behavior** of value baskets,
- Operates **conditionally on macro regimes**, &
- Uses **descriptive comparisons**, not predictions.

## What VinV Is For

- Understanding **durability, dispersion**, & **recovery** characteristics,
- Translating **macro diagnostics** into **observed asset behavior**, &
- Demonstrating **applied execution capacity**.

*VinV does not forecast, time, optimize, or recommend allocation.*

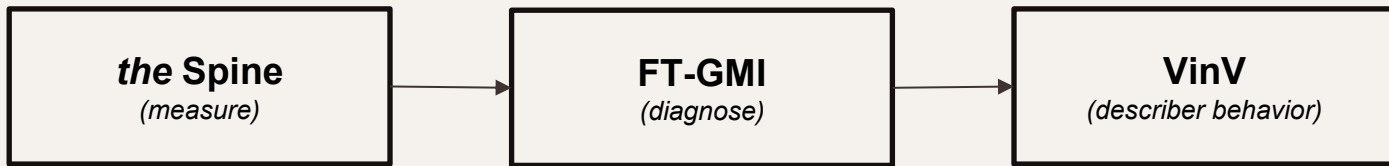
# Boundary

## Boundary Rules

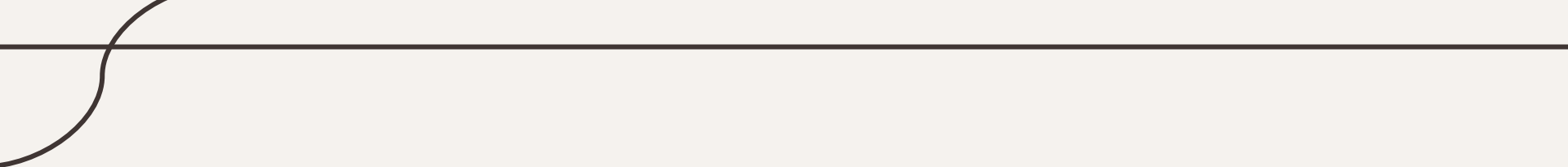
- VinV consumes upstream outputs **read-only**,
- VinV **cannot** tune, validate, or alter FT-GMI, &
- VinV success or failure **does not affect** diagnostics.

## Supervisory Implication

- **No forecasting oversight** requested,
- **No performance endorsement** implied, &
- **No** expansion of supervisory scope.



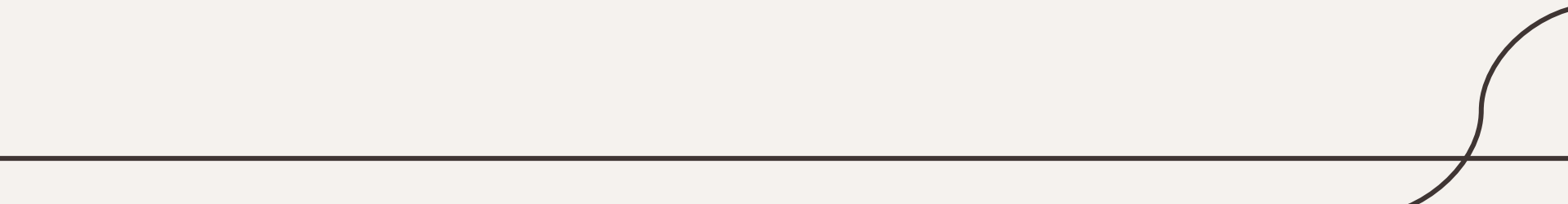
*VinV exists to show applied execution, not to extend diagnostic authority.*



# 02

# Existing Data Artifacts

*Documents the governed data artifacts already constructed, independent of modeling or forecasting assumptions.*





# Existing Data Artifacts (Overview)

*What Exists Today (Concrete, Not Aspirational)*

## Artifacts in place

- **Canonical equity-macro panel,**
- **Returns integration & validation pipeline,**
- **Regime-aware temporal structure, &**
- **Defensive data contracts & checks.**

*These artifacts exist independently of any model choice or forecasting logic.*

# Canonical Equity–Macro Panel

*Long-Horizon, Regime-Compatible Panel*

## Panel characteristics

- **Monthly frequency, symbol–date indexed,**
- Clean **equity identifiers & returns,**
- **Time-aligned macro variables, &**
- **Stable schema & versioned inputs.**

## Design intent

- Enable **cross-sectional comparisons,**
- Support **regime segmentation, &**
- Avoid ad-hoc feature construction.

*No raw provider data enters VinV directly.*

# Returns Integration & Validation

*Returns Are Explicitly Governed*

## What is implemented

- **Normalized monthly returns** (*ret\_m*),
- **Column presence & type validation**,
- **Defensive joins** (*no silent drops*), &
- **Symbol/date sorting & integrity checks.**

## Why this matters

- **Eliminates common backtest leakage**,
- Makes **downstream evaluation auditable**, &
- **Separates data hygiene** from **modeling**.

# Feature Construction Layer

*Descriptive, Cross-Sectional Features*

## Feature families

- **Rank-based measures,**
- **Z-score normalization,**
- **Dispersion & breadth metrics, &**
- **Persistence & churn indicators.**

*Features describe relative structure, not expected payoff.*

# Regime-Aware Evaluation Windows

*Time Is Treated as Conditional*

## What exists

- **Regime-segmented analysis windows** defined upstream by FT-GMI,
- **Explicit stress-period slices**, &
- **Post-stress recovery windows**.

## Why

- **Avoids full-sample averaging**,
- Treats **crises as information, not noise**, &
- Aligns with **applied macro time-series logic**.

# Artifact Independence (Boundary Reinforcement)

*Artifacts Are Upstream-Stable*

## Key property

- All **VinV artifacts remain valid** if:
  - **FT-GMI outputs** are **removed**,
  - **Models** are **changed**, &
  - **Forecasting** is **never added**.

*The data foundation does not assume, require, or imply prediction.*



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# 03

# Regime Conditioning

*Positions macro regimes as contextual states for interpretation rather than objects of prediction or timing.*



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# Regime Conditioning (Why It Exists)

*Why Regime Conditioning Is Non-Optional*

## Core premise

Macro-financial behavior is **state-dependent**, not stable across time.

## Design stance

- **Full-sample averages** are **misleading**,
- **Structural breaks** are **information**, not noise, &
- **Stress periods** are **analytically distinct**.

*VinV treats regime shifts as first-class conditions, not exceptions.*



# What a “Regime” Means Here

*Regimes Are Contextual, Not Predictive*

## Regime definition *(operational)*

- A regime is a **persistent macro environment** characterized by:
  - **Constraint tightness,**
  - **Liquidity conditions, &**
  - **Stress transmission intensity.**

## Clarification

- Regimes are **not forecasts,**
- Regimes are **not labels for timing, &**
- Regimes are used solely to **condition observation.**

*Regime identification does not imply regime transition prediction.*

# Temporal Segmentation Logic

*How Time Is Treated*

## Segmentation approach

- **Long-run baseline context,**
- **Financialization era context, &**
- **Stress-intensive sub-windows.**

## Why this matters

- **Prevents dilution of crisis behavior,**
- **Avoids smoothing away nonlinear responses,**  
&
- **Preserves interpretability under stress.**

*Time is segmented to observe behavior, not to optimize outcomes.*

# Conditional Evaluation (Not Pooling)

*What Conditioning Enables*

## With regime conditioning

- **Like-for-like comparisons,**
- **Cross-sectional behavior** under **shared constraints,** &
- **Meaningful stress** vs **non-stress contrasts.**

## Without conditioning

- **False stability,**
- **Misleading averages,** &
- **Overconfident conclusions.**

*VinV does not pool incompatible macro environments.*

# Relationship to Diagnostics (Boundary Reinforced)

*How This Relates to FT-GMI*

## Clarification

- **FT-GMI diagnoses macro structure,**
- **VinV conditions on macro context,**
- **No diagnostic output is altered by VinV, &**
- **Diagnostic immutability is enforced by design,** not convention.

*Conditioning uses macro context as a lens, not as a target.*

# Why This Matters for Interpretation

## *Interpretive Discipline*

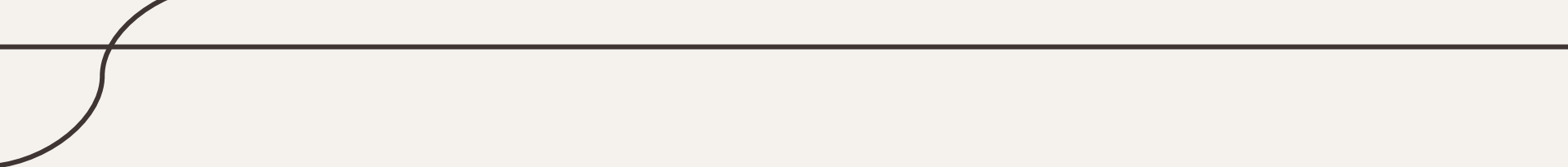
### Regime conditioning allows

- **Honest statements** such as:
  - “*Behavior differs under stress*”, &
  - “*Recovery paths are regime-specific.*”

### What it prevents

- **Universal claims,**
- **Overgeneralization, &**
- **Implied prediction.**

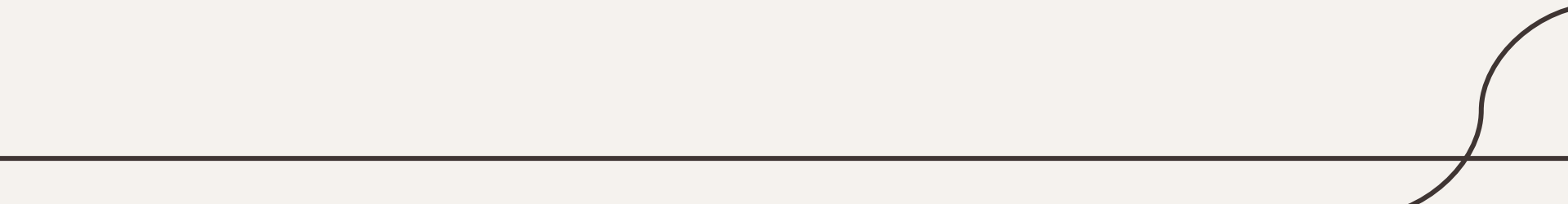
*Regime conditioning narrows claims to what the data can support.*



04

# Behavior Under Stress

*Examines how equity structures behave under macro stress without asserting performance, signals, or tradability.*



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# Global Comment

*(Applies to ALL Graphs in This Section)*

All figures are **descriptive, regime-conditional comparisons** of VinV relative to standard benchmarks during identified macro stress windows.

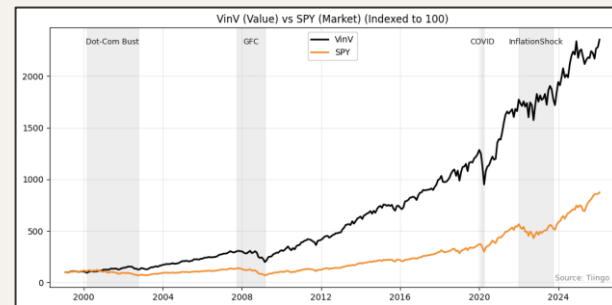
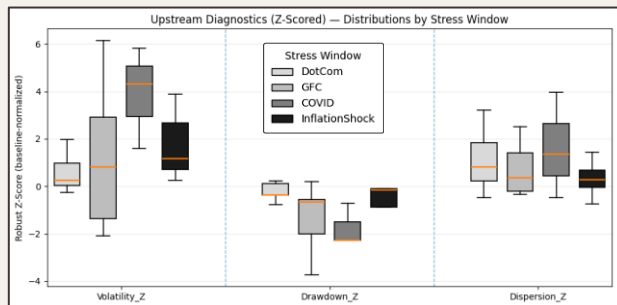
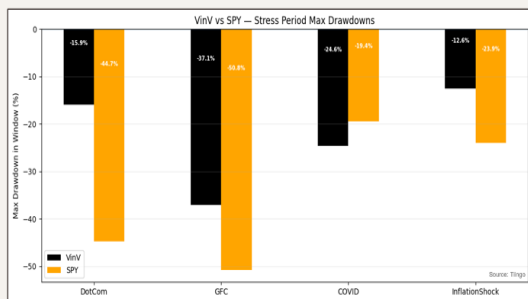
Graphs are **indexed to a common starting point, conditioned on shared stress regimes**, & intended to illustrate **structural behavior (drawdown, volatility, dispersion, recovery shape)**, **not prediction, timing, allocation, or performance claims**.

All stress windows shown correspond to **historically recognized U.S. macro dislocations** (*Dot-Com, GFC, COVID, Inflation Shock*).

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# Global Comment – Stress Periods

*Stress periods are inputs, not conclusions.*



| Stress Period                 | Dates               | Why This Period Qualifies as Stress (Minimal, Factual)                                                    | What FT-GMI Uses It For                                             |
|-------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| DotCom Bust                   | Mar 2000 – Oct 2002 | <b>Equity</b> valuation <b>collapse</b> following capital misallocation & tightening financial conditions | <b>Regime contrast, equity fragility, recovery asymmetry</b>        |
| Global Financial Crisis (GFC) | Jul 2007 – Mar 2009 | Systemic credit contraction, liquidity seizure, <b>synchronized global drawdown</b>                       | <b>Liquidity transmission, policy constraint diagnostics</b>        |
| COVID Shock                   | Feb 2020 – Apr 2020 | Sudden <b>exogenous halt</b> in economic activity with extreme volatility & liquidity stress              | <b>Shock absorption vs amplification behavior</b>                   |
| InflationShock                | Jan 2022 – Oct 2023 | <b>Inflation regime shift</b> with aggressive global monetary tightening & valuation compression          | <b>Inflation constraint regimes, cross-asset stress interaction</b> |

*Stress period definitions are fixed upstream by FT-GMI and consumed read-only by VinV.*



# VinV vs SPY –

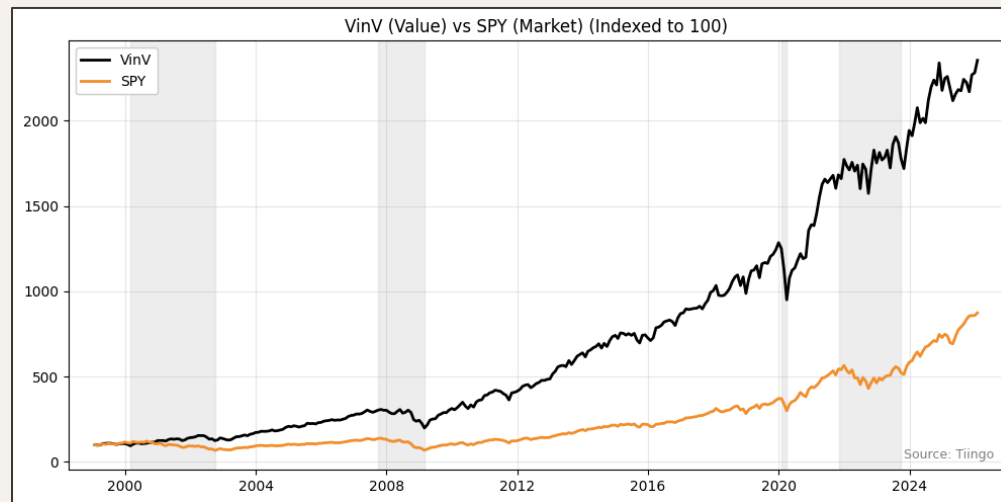
*Long-Horizon Capital Trajectories (Indexed)*

## Comment

- **VinV** reflects the aggregated capital path of **dividend-persistent equities** constructed under **fixed inclusion rules**, &
- **SPY** serves as a **broad-market reference** for **regime-level comparison**, not **performance benchmarking**.

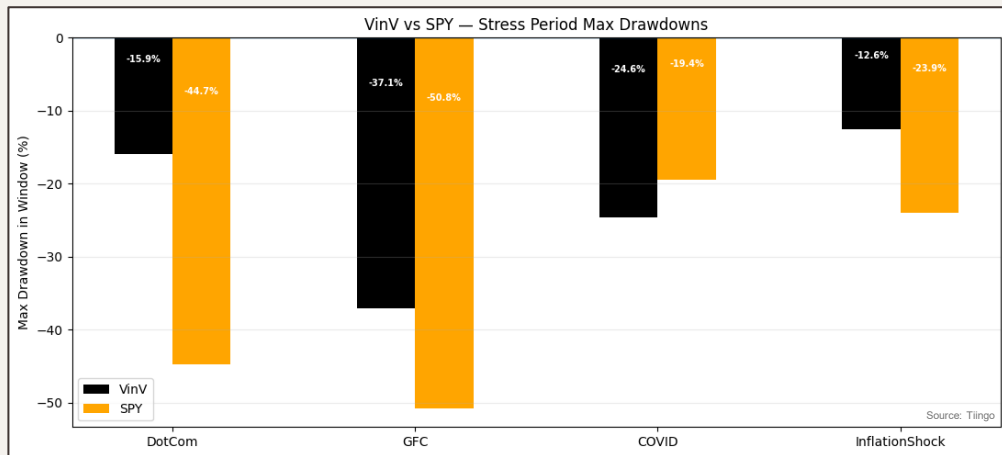
## What it is NOT

- **Not** an **outperformance** or **alpha comparison**,
- **Not** a **return forecast** or **expectation**, &
- **Not** a **trading** or **allocation signal**.



# VinV vs SPY —

*Maximum Drawdowns Across Major Stress Regimes*



## Comment

- **VinV** exhibits **materially different drawdown depth** across **stress regimes** relative to the **broad market**, reflecting **structural composition** rather than **defensive positioning**, &
- The comparison **isolates loss severity** within **defined stress windows**, **independent of recovery timing** or **post-crisis performance**.

## What it is NOT

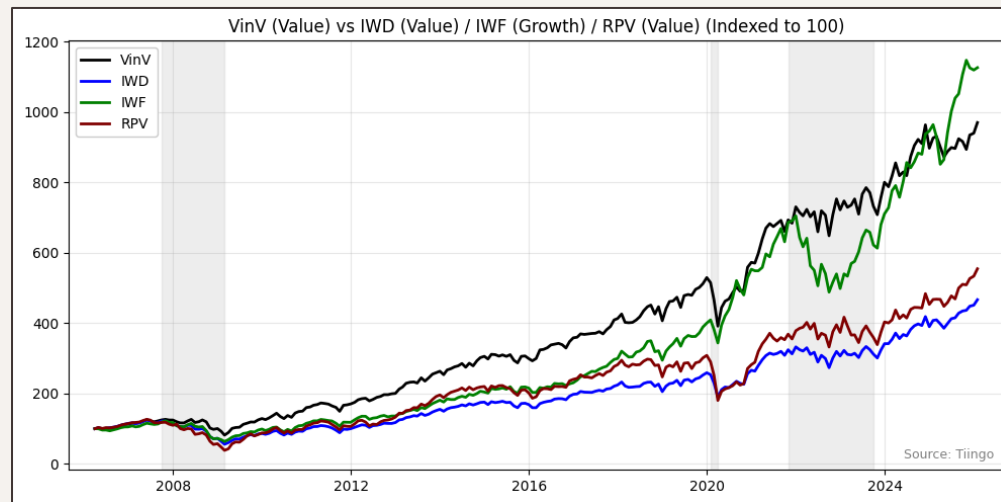
- **Not** a **recovery timing** tool,
- **Not** a **forecast of normalization speed** or **policy response**, &
- **Not** a **fitted model** or **optimized portfolio**.

# VinV vs Benchmarks –

*Long-Horizon Capital Paths Across Value & Growth Constructions (Indexed)*

## Comment

- **VinV** reflects the **long-horizon capital path** of a **fixed, value-oriented equity construction** evaluated under **identical market history** as **standard style benchmarks**, &
- **IWD, IWF, & RPV** serve as **reference constructions** representing **conventional value & growth exposures**, included to **contextualize structural differences in compounding behavior** rather than to benchmark performance.

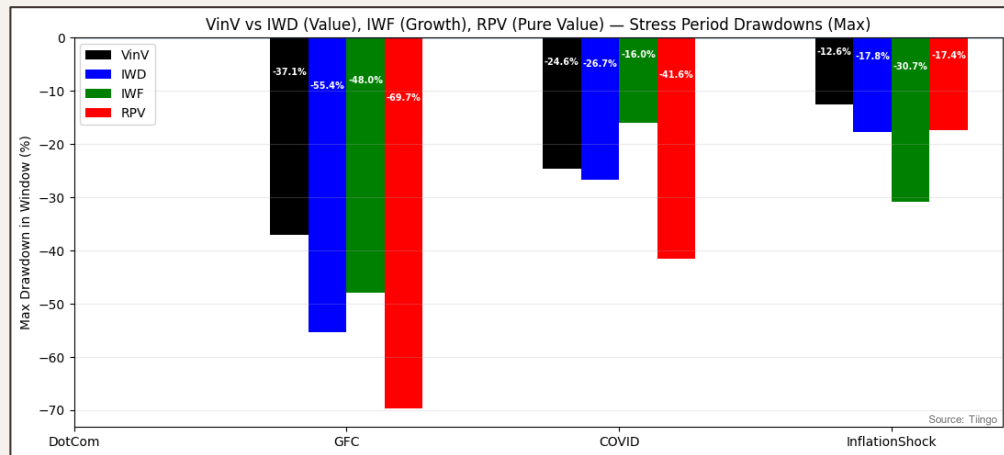


## What it is NOT

- **Not** an **outperformance, alpha, or factor dominance** claim.
- **Not** a **return forecast, expected payoff, or capital allocation recommendation**, &
- **Not** evidence of **robustness outside the defined constructions or historical window shown**.

# VinV vs Benchmarks –

*Relative Drawdown Comparison (Showing Depth & Period)*



## Comment

- Across stress regimes, VinV exhibits a **distinct drawdown profile** relative to **style benchmarks**, reflecting **portfolio construction** & **factor exposure** rather than defensive positioning or active risk mgmt., &
- Observed differences are **regime-specific** & do not imply consistency of behavior outside defined stress windows.

## Why this is allowed

- Drawdown is treated as a risk characteristic, not a payoff.

# Stress Window Diagnostic Scorecard –

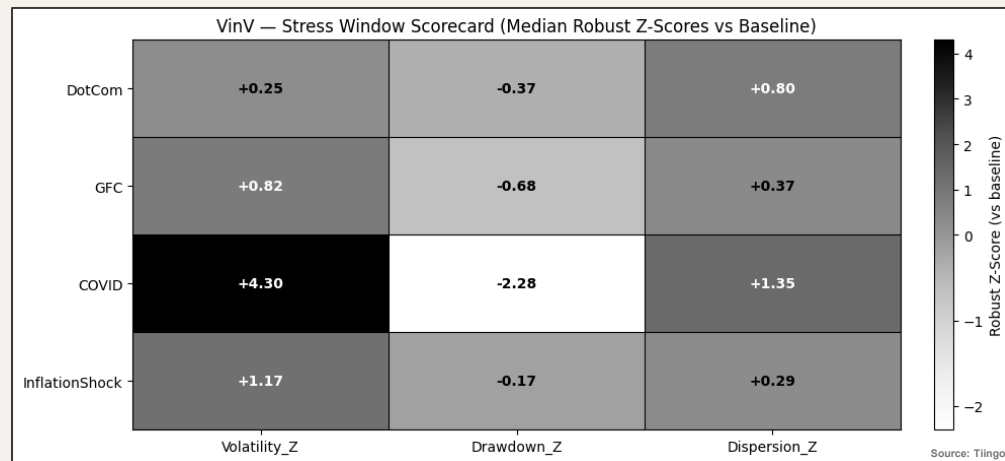
*Median Robust Z-Scores vs Long-Run Baseline (Regime-Conditioned)*

## Comment

- **Summarizes VinV's stress behavior across regimes** using volatility, drawdown, & dispersion deviations from baseline.
- **Uses robust Z-scores (median / IQR)** to remain valid under non-Gaussian stress conditions.
- **Enables regime-specific comparison** without averaging across episodes.

## What it is NOT

- **Not a performance or alpha ranking.**
- **Not a forecasting, timing, or allocation signal.**
- **Not a causal claim** about macro drivers.



**Diagnostics shown** as robust Z-scores (median / IQR), normalized to long-run baseline; **volatility** reflects rolling dispersion, **drawdown** reflects rolling peak-to-trough loss, & **dispersion** reflects cross-sectional spread.

# VinV – Methodological Guardrails

*(Non-Negotiable)*

## Comment

- **VinV** results shown here are **descriptive** & **conditional** on the **specified windows** & **definitions**.
- **Outputs** are **not** used to generate **forecasts**, **timing rules**, or **allocation directives** in this deck.

## What it is NOT

- Not an **allocation engine**
- Not an **optimization workflow**, &
- Not **causal attribution** (no “because” language).

*All comparisons are observational; downstream extensions are explicitly deferred.*

# Behavioral Summary Slide

## Permitted summary language

- **Stress impacts** are **regime-dependent**,
- **Drawdown & recovery paths differ structurally**, &
- **Volatility & dispersion respond asymmetrically**.

## Explicit non-claims

- **No dominance**,
- **No stable ranking**, &
- **No predictive inference**.

# VinV – Deferred Completion Path

*(Post FT-GMI Freeze)*

## Comment

- VinV is intentionally presented here as a **bounded proof-of-concept supporting FT-GMI credibility** via **execution discipline**.
- After **FT-GMI** is **frozen**, VinV will expand into a separate, explicitly **forecast-oriented track** with dedicated **validation & governance**.

## What it is NOT

- **Not** a claim that these steps are **completed**,
- **Not** a **performance-marketed** product, &
- **Not** an **implied trading system**.

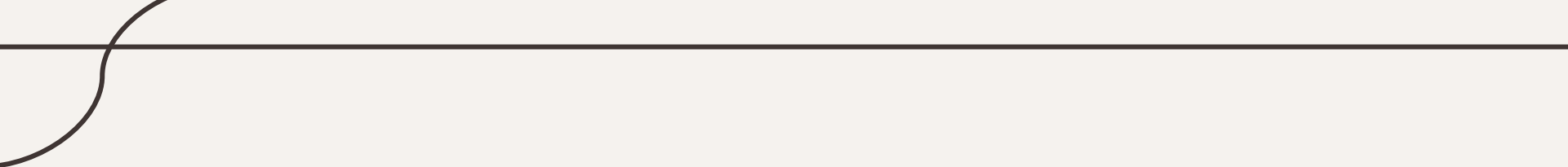
## Planned

*(Not Yet Implemented)*

- **Benchmark suite freeze & reproducibility manifest**,
- **Walk-forward evaluation** design (**out-of-sample**),
- **Factor attribution & sensitivity analysis**, &
- Reporting layer appropriate for **quant research roles**.

*All comparisons are observational; downstream extensions are explicitly deferred.*

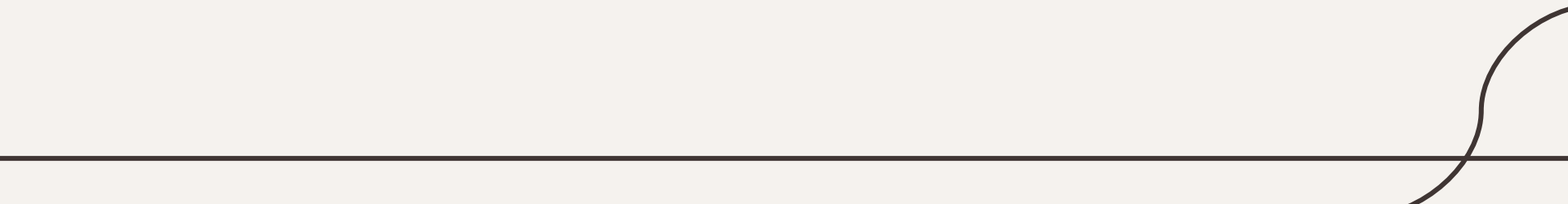




05

# Observed Structure

*Characterizes post-stress recovery shape & persistence as descriptive path properties,  
not indicators of timing, signals, or allocation.*



# Observed Structure (Bounded Summary)

## *Observed Structure: Descriptive Patterns Only*

### **Purpose**

Summarize **recurring structural characteristics** observed across stress regimes **without inference, ranking, or prediction**.

### Observed properties

- **Regime dependence** — behavior varies materially across macro environments,
- **Asymmetric drawdowns** — depth & duration differ across constructions,
- **Nonlinear recovery paths** — recoveries are uneven & path-dependent, &
- **Volatility clustering** — instability concentrates during stress regimes.

*These observations describe how structures behaved, not why or what comes next.*

# Interpretive Limits (Explicit Non-Claims)

*What Is Intentionally Not Inferred*

## From the Observed Structure

VinV **does not infer**:

- **Causality,**
- **Predictive signals,**
- **Stable rankings or dominance,**
- **Allocation or timing implications, &**
- **Regime transition forecasts.**

## Reason

These inferences exceed what descriptive, regime-conditional evidence can support.

*Observed structure represents the ceiling of responsible interpretation at this stage.*



# 06

# Completion Path

*Synthesizes observed stress & recovery behaviors into a bounded empirical profile, explicitly defining interpretive limits & excluding forecasting, signaling, or allocation claims.*



# Completion Path (Section Framing)

*Completion Path: Sequencing, Not Expansion*

## Core premise

- **VinV** is **intentionally incomplete** while **FT-GMI** is being finalized, &
- This is a **sequencing decision, not a capability gap**.

*Completion occurs by unlocking downstream work, not by revisiting upstream diagnostics.*

# What “Completion” Means for VinV

*Definition of VinV Complete (Applied, Not Marketed)*

## Completion

**VinV** is considered **complete** when it includes:

- **Fixed feature contracts** from **FT-GMI** & **the Spine**,
- **Finalized regime segmentation**,
- **Fully specified evaluation windows**,
- **Documented forecasting & benchmarking logic**, &
- **Reproducible results artifacts**.

*Completion does not change FT-GMI or its diagnostics.*

# Why FT-GMI Must Be Fixed First (Diagnostic Immutability)

*Upstream Stability Enables Downstream Rigor*

## FT-GMI finalization provides

- **Stable macro diagnostic definitions,**
- **Locked regime context, &**
- **Non-moving feature semantics.**

## Why this matters

- **Prevents re-training on shifting definitions,**
- **Eliminates silent leakage, &**
- **Preserves interpretability.**

*Diagnostic immutability is a prerequisite for applied evaluation.*

# What Unlocks Once FT-GMI Is Final

*Downstream Additions (Explicitly Deferred)*

## Downstream Additions (Explicitly Deferred)

Once FT-GMI is finalized, VinV adds:

- **Forecasting models** (*supervised; statistical & ML methods explicitly scoped*),
- **Benchmark comparisons** (explicit baselines),
- **Cross-regime validation,**
- **Stress-aware performance evaluation, &**
- **Applied research reporting artifacts.**

*These additions occur entirely downstream & do not alter diagnostics.*



# Why This Sequencing Is Deliberate

## *Risk Management by Design*

### This sequencing prevents

- **Back-propagation** of outcomes into diagnostics,
- **Overfitting** to regime definitions, &
- **Premature performance claims.**

### This sequencing enables

- **Cleaner empirical interpretation,**
- **Clear academic boundaries, &**
- **Stronger applied credibility** later.

# Supervisor Impact (Explicitly Limited)

*Why This Does Not Expand Supervisory Scope*

## No supervision requested for

- **Forecasting choices,**
- **Model selection,**
- **Performance evaluation, &**
- **Optimization or trading logic.**

## Supervisory role remains

- **Macro framing,**
- **Regime logic, &**
- **Stress interpretation sanity checks.**

*VinV completion does not add supervisory burden.*

# End-State Snapshot

*End State (Clean Separation Preserved)*

## Properties of the end state

- **Diagnostics** remain **non-forecasting**,
- **Application layer** becomes **fully specified**, &
- **No feedback loops introduced.**



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# Why This Path Matters

*By fixing diagnostics first & completing application second, VinV becomes empirically strong without compromising academic integrity.*

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# Thank You

**Rand Sobczak Jr.**

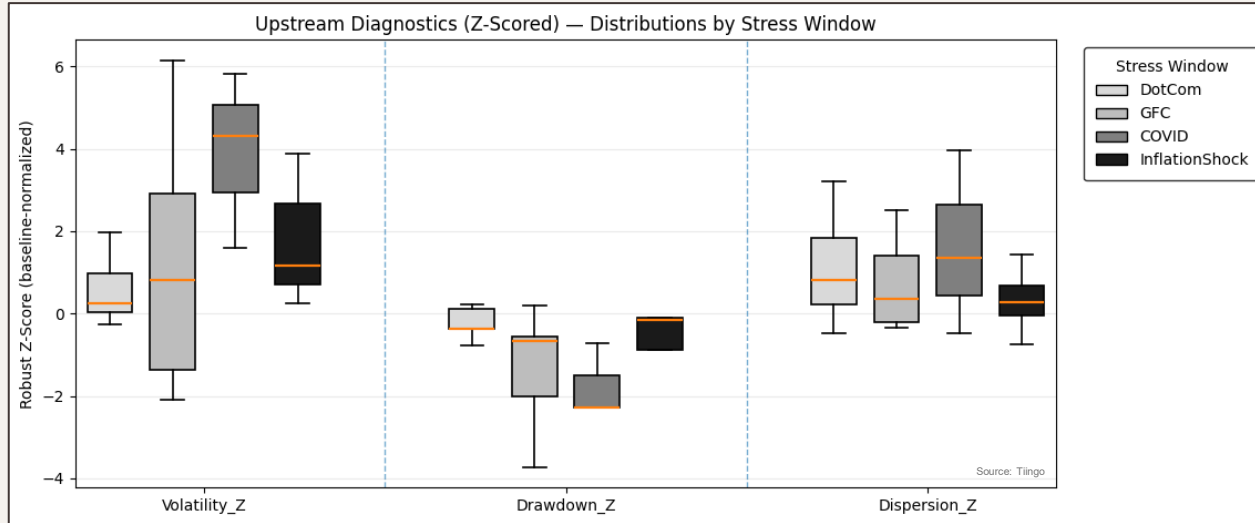
**M.S. Data Science** | University of Pittsburgh, *2026 exp.*

**B.A. Economics** | Michigan State University



# Appendix: Z-Score Distributions by Regime

## *Robust Diagnostics Under Stress (Baseline-Normalized)*



**Diagnostics** as Robust Z-scores (median / IQR), normalized to long-run baseline;  
**Volatility** for rolling dispersion, **Drawdown** for rolling peak-to-trough loss, &  
**Dispersion** for cross-sectional spread.

*Distributional diagnostics confirm regime-level medians shown in the main heatmap are representative, while preserving visibility into dispersion, skew, & tail behavior in stress regimes.*